

MB502T - Molecular Biology (Final Term-2024)

Function of A site, P site and E site:

The A site is where new amino acids are added to the growing chain, the P site is where the chain is transferred, and the E site is where the empty tRNA exists.

Classes of Splicing: (3 classes)

- 1- Nuclear pre-mRNA splicing
- 2- Group II introns splicing
- 3- Group I introns splicing.

What is plasmid and where it is present

A plasmid is a small, circular DNA found in the cytoplasm of bacteria and some other organisms. It carries extra genetic info and can replicate independently. It is playing a role in gene transfer, antibiotic resistance, and genetic engineering.

Name 2 protein necessary for lactose mechanism:

- 1- β -galactosidase
- 2- Lactose Permease

Initiation of transcription in Prokaryotes:

In Prokaryotes, transcription initiation involves RNA polymerase ^{binds, & unwinds the DNA.} recognizes the DNA promoter. The sigma factor helps in promoter recognition. RNA polymerase synthesizes RNA complementary to DNA strands forming mRNA transcript.

Alternative Splicing and its important:

Alternative Splicing allows one pre-mRNA to create different mRNA molecules by including or excluding specific parts. This increases genetic complexity, functional diversity and adaptability in organisms playing a crucial role in cellular specialization & disease development.

CTD heptatide in Yeast, human and Drosophila:

In Yeast, There are 27 CTD heptatide repeats, in *Caenorhabditis elegans* there are 32, in *Drosophila* there are 45, and in humans there are 52.

Initiation in Prokaryotes:

OR Initiation of Transcription in Prokaryotes OR
How transition to open complex are worked?

To transition to an open complex in transcription, RNA polymerase binds to the promoter region & undergoes conformational changes. The closed complex form initially, then local DNA unwinding occurs, creating the open complex.

This allows for RNA synthesis to begin.

^{adenosine deaminase acting on RNA.}
ADAR: Name 3 genes codes ADAR in human:

ADAR is an enzyme that converts adenosine to inosine in RNA, affecting gene expression. In humans, there are 3 genes that codes for ADAR: ADAR 1, ADAR 2 and ADAR 3.

These enzymes play roles in RNA editing, impacting protein diversity and nervous system function, and are linked to neurodevelopmental disorders and cancers.

Topoisomerase:

is an enzyme that alters the topology of DNA by adding or removing twists and knots. It helps with DNA replication, transcription and repair, while preventing excessive coiling.

Why plasmids are not regarded as cell's genome:

Plasmids are not regarded as parts of cell's genome for 2 reasons
1- Plasmids can be found in different species and can move b/w them. 2- Plasmids may be present or absent in cells of the same species and are not necessary for cell growth & division.

Why mRNA has short lifespan:

because enzymes called ribonucleases break it down. This helps cells quickly regulate gene expression and adapt to changing conditions.

Aflatoxin B1:

produced by *A. flavus*, is a powerful toxin that can damage DNA, leading to mutations and health problems. It's crucial to control aflatoxin exposure to prevent these issues.

RNA Processing Events:

RNA Processing modifies primary transcripts. In eukaryotes, pre-mRNA undergoes splicing to remove introns & join exons. A 5' cap and a poly-A tail are added for stability. These modifications ensure stable, transportable mRNA and accurate protein synthesis.

Five features of Operon Model:

- 1- Genes combined in one mRNA.
- 2- Promoter near lac region
- 3- Operator binds repressor
- 4- Repressor block transcription.
- 5- Inducers release repressor, allowing transcription

Promoter, Oligomer, Multimer:

- * A promoter is a DNA sequence that starts gene expression.
 - * An oligomer is a short chain of repeating molecules.
 - * A multimer is a complex made up of multiple identical or similar subunits.
- OR ^{Change in} Importance ^{affects} of nucleosome in transcription machinery:

because they help regulate DNA accessibility. They can be moved to allow transcription factors and RNA polymerase to interact with DNA, promoting gene expression while preserving chromatin structure.

End Replication Problem:

The end replication problem occurs when RNA primers are needed to start DNA synthesis at the ends of linear chromosomes. This creates a dilemma for replication.

Types of Splicing:

There are two types of Splicing;

- 1- Constitutive Splicing: joins exons and remove introns.
- 2- Alternative Splicing: selectively includes or excludes exons, creating diverse mRNA isoforms.

↓ یہاں سے لکھنا شروع کریں۔

CPD & its function:

A Cyclobutane Pyrimidine Dimer (CPD) is a structure formed by covalent bonding of two adjacent pyrimidine bases (thymine or cytosine) in DNA due to UV radiation.

CPD plays a function in inducing DNA repair mechanisms and help repair UV-induced DNA damage.

Semi-Conservative model of DNA replication:

The semi-conservative model of DNA replication says that each parent strand act as a template for a new strand resulting in two daughter DNA molecules with one original and one newly synthesized strand.

Primer & its function:

is a short RNA or DNA sequence (typically 18-25 nucleotides long) that initiates DNA synthesis during replication or PCR. It provides starting point for DNA polymerase to add complementary nucleotides, enabling the synthesis of new DNA strand.

Operon Model:

In this model, a cluster of genes is transcribed together to form a single mRNA molecule. This mRNA encodes multiple proteins. This operon consists of an operator, promoter, regulator, and structural genes. Repressor^{protein} blocks transcription.

RNA editing:

is a process where cells can make changes to specific nucleotide sequences in RNA after it's been generated. It includes inserting, deleting or substituting nucleotides in the RNA molecule.

2 protein in E. coli that necessary for lactose metabolism:

1-cAMP

2-CAP (Catabolite activator protein)

Involvement of CTD tail in Transcription:

The CTD tail of RNA polymerase II is important (in^x transcription) for transitioning from initiation to elongation during transcription. It undergoes Ser2 phosphorylation, a key regulatory step.

Clustered
Regularly
Interspaced Short
Palindromic
Repeats

CRISPR & its application:

CRISPR is a powerful tool for editing genomes. It allows researchers to modify DNA sequences and gene functions, correct genetic defects, treat diseases, and improve crops.

Galactose Operon:

The gal operon in prokaryotes encodes ^{enzyme} for galactose metabolism. It has two operators, O_E ^{external} and O_I ^{internal}, and consists of three structural genes: $galE$ ^(epimerase), $galT$ ^(transferase) and $galK$ ^(kinase). These genes are transcribed from promoter P_{G1} and P_{G2} upstream from $galE$.

Spliceosome:

is a complex molecular machine involved in ~~m~~ RNA splicing, that removes the introns and joins exons in pre-mRNA.

Crucial role in gene expression & production of mature mRNA ^{mole} ^{cule}

Dalgaard Sequence (Shine Dalgarno Sequence):

is a nucleotide sequence found in bacterial mRNA that helps in the initiation of protein translation. It interacts with ribosomes to ensure proper positioning for translation to occur.

Allostery:

is a phenomenon where a molecule binds to a protein and affects its activity at different site. It's like a remote control that can change the behavior of proteins. e.g binding of O_2 to hemoglobin.

PUC Plasmid:

also known as the PUC vector, is a commonly used cloning vector in molecular biology research. It contains feature such as a multiple cloning site (MCS) and selectable markers that make it easy to insert and propagate DNA fragments of interest. It's great for genetic engineering.

Torpedo Model:

It refers to the mechanism where an enzyme degrades the second RNA strand as it comes out of polymerase, which may lead to termination. It's called the Torpedo model of termination, that stops RNA synthesis process.

Function of Lambda phage:

is used in gene library ~~for~~ production. It enters E. coli cells efficiently and can integrate large DNA fragments.

To clone long DNA fragments, the phage genome's lysogenic life cycle coding is replaced with foreign DNA.

Riboswitches: e.g. *Bacillus subtilis*

Riboswitches control gene expression based on small molecule conc.

They bind ligands and change structure, impacting transcription or translocation. They are found near genes related to the ligand.

Charged and Uncharged Amino acids:

tRNA molecules to which an amino acid is attached are said to be Charged, and tRNA that lack an amino acid are said to be uncharged amino acid.

Steps of Attachments of amino acid to tRNA:

Enzyme recognizes specific amino acid and tRNA.

Enzyme transfers amino acid from ATP to tRNA.

Amino acid attaches to tRNA through ester bond.

Charged tRNA is ready for protein synthesis.

These steps ensure the right amino acid is carried by tRNA during production protein.

Cosmid:

is a hybrid plasmid used for cloning large DNA fragments in gene libraries.

Catabolic Repressor:

is a protein that controls gene expression by binding to DNA. It stops gene expression when substrates are absent and allows it when substrates are present.

Transcriptional Repressor:

is a protein that stops gene expression by binding to DNA.

Act as negative regulators, inhibiting gene transcription.

Two 5'-3' RNA modifications:

Two common 5'-3' RNA modification are the addition of a 5' cap and a poly (A) tail. The 5' cap helps protect the RNA and the poly (A) tail plays a role in stability and translation.

Elongation Phase in Eukaryotes:

During elongation phase in eukaryotes, RNA polymerase II continues to transcribe the DNA template into RNA, adding nucleotides to the growing RNA strand. This phase is important for the synthesis of the full-length RNA molecule.

Finding of Franklin and Wilkins:

Franklin and Wilkins were scientists who made significant contributions to the field of DNA structure. F and W discovered the double helix structure of DNA through crystallography.

How CPD can alter DNA function and structure:

CPDs, caused by UV radiation, can change DNA structure and function by forming covalent bonds b/w bases. This can disrupt replication and transcription, leading to mutations and potential disease.

How excision base repair is utilized in which types of DNA damage:

EBR is utilized in various types of DNA damage, including UV-induced pyrimidine dimers and chemical modifications such as alkylations and oxidations. It helps to remove and replace damaged bases, maintaining the integrity of DNA sequence.

Two functions of general transcription factors:

⇒ They help in the binding of RNA polymerase to the promoter region of the DNA. ⇒ They assist in the initiation of transcription by helping to unwind the DNA strands.

Euchromatin and Heterochromatin:

Euchromatin: less condensed & more accessible for gene expression

Heterochromatin: highly condensed & typically associated with gene repression.

Function of releasing factor:

is to recognize the stop codon during translation and releasing the newly synthesized protein from the ribosome.

SR Proteins:

SR proteins, or serine/arginine-rich proteins, are a family of RNA-binding proteins involved in RNA splicing, a crucial step in gene expression. They play a role in regulating alternative splicing by binding to specific RNA sequences and facilitating the assembly of the spliceosome. This process contributes to the generation of diverse mRNA isoforms from a single gene, influencing protein diversity and function. OR

are RNA binding proteins that regulate RNA splicing. They help creating different mRNA isoforms from a single gene, influencing protein diversity.

Effect of oxygen species produced in cellular respiration on DNA functions

Oxygen species produced during cellular respiration can have
have damaging effects on DNA. * Cause oxidative stress and
lead to DNA damage, such as DNA strand breaks & modifications.
* Potentially causing mutations.

Telomeres:

are protective caps at the ends of chromosomes that preserve
our genetic material. They shorten as cells divide, contributing to
^{How} Gene Expression is controlled by regulatory Proteins: ^{Cell aging.}

Regulatory Proteins control gene expressing by binding to DNA
and influencing transcription. They activate or repress genes, regulating
their expression.

Function of class II release factor:

Class II release factors in translation recognize the stop codon
on mRNA and promote the termination of protein synthesis by
newly synthesized proteins from the ribosome.

Self Splicing:

is when certain RNA molecules can remove their own non-
coding regions without help.

Open Reading frame:

is DNA or RNA sequence that can be translated into a
protein. It starts with a start codon (usually AUG) and ends with
a stop codon (such as UAA, UAG or UGA). ORFs are important in
identifying potential genes & studying gene expression.

How mRNA is transported from nucleus to cytoplasm:

through a process called mRNA export. This complex help
transport mRNA through nuclear pores, which are large
protein channel in the nuclear envelop.

GAGA binding factor:

is a protein that binds to specific DNA sequences and plays a role in gene ~~expression~~ regulation. * helps regulate gene expression.
* influencing the structure of chromatin.

Function of Sigma factor: in bacteria:

Sigma factor in bacteria help RNA polymerase to initiate transcription by guiding it to specific DNA ~~plaf~~ promoter region.

Base Excision Repair: (BER) ② ✱

Base excision repair is a DNA repair mechanism that corrects small, non-bulky lesions in the DNA, such as modified or damaged bases. It involves the removal of the damaged base by specific enzyme (glycosylase), followed by the replacement of the correct base through DNA synthesis (polymerase) and ligation.

✓ Three Functions of Nucleic Acids: (7) ★

Nucleic acids have a very diverse set of functions, such as;
* cell creation * the storage (of) and processing of genetic information
* Protein building * and the generation of energy cells.

✓ CPD Photolyase and its Functions: (8) ★

⇒ CPD Photolyase is a protein that helps repair cyclobutane pyrimidine dimers (CPDs) in DNA.

⇒ It has a unique structure that allows it to absorb light in the visible spectrum, specifically blue light.

⇒ When CPD ~~big~~ photolyase absorb light, it undergoes a series of chemical reactions that enable it to break the bonds within the CPD and restore the DNA to its original state.

⇒ It's like a DNA repair that uses the power of light to fix

those ↓ dimers.
pesky

Chargaff's Rule: * (20)

was discovered by Erwin Chargaff in 1940s. Chargaff's rule also known as base pairing rule, states that in DNA, the amount of adenine (A) is equals to the amount of thymine (T), and the amount of guanine (G) is equals to the amount of cytosine (C), forming complementary base pairs.

Watson And Crick Model: (13) ★

The Watson and Crick Model for DNA, proposed in 1953, describes the structure of DNA as a double helix with two anti-parallel strands linked by hydrogen bonds b/w complementary base pairs (adenine-thymine and guanine-cytosine). The sugar-phosphate backbones run on the outside, while the bases are stacked inside. This model provided the foundation for understanding DNA replication, transcription, and genetic information storage.

Primers and its Functions: (14) ★

- ⇒ Primers are short-stretches of DNA, typically 18 to 25 nucleotides long, target unique genome sequences.
- ⇒ Synthesized in lab, and are used in many different ways by binding to the template through complementary base pairs.
- ⇒ Taq polymerase, like other DNA polymerases, relies on primers to initiate DNA synthesis.

Three ways of Base Excision Repair (BER): (19) *

- 1- Glycolyase-mediated BER: DNA glycolyases recognize and remove damaged bases.
- 2- AP endonuclease mediated BER: AP endonucleases cleaves the DNA backbone at sites lacking a base.
- 3- DNA polymerase and ligase-mediated BER: DNA polymerase fills the gap with the correct nucleotides, and ligase seals the nick in the DNA backbone.

✓ Telomerase And it's functions:

(31) *

* Telomerase is an enzyme that helps maintain the length of telomeres, which are protective caps on the ends of our chromosomes.

* It adds repetitive DNA sequences to the telomeres, preventing them from getting shorter during cell division.

* This helps preserve the integrity of our genetic material.

✓ Why C-N bond of peptide can't rotate freely: (32) *

⇒ The C-N bond in a peptide cannot rotate freely because of a concept called partial double bond character.

⇒ This means that the C-N bond has a certain amount of double bond character, which restricts its rotation.

✓ Nucleotide Excision Repair (1973): Five Steps: (33) ★

⇒ NER is a DNA repair mechanism that removes bulky DNA lesions

⇒ The five steps of NER:

1- Lesion Recognition

2- Damage Verification

3- Dual Incision

4- Excision of the damaged DNA fragments

5- DNA synthesis and ligation to restore the original DNA sequence.

✓ How phosphodiester bond is formed: (34) ★

⇒ Phosphodiester bonds are formed during DNA and RNA synthesis through a condensation reaction b/w the 3' hydroxyl (OH) group of one nucleotide and the 5' phosphate group^(-PO₄) of another nucleotide.

⇒ This reaction releases a water molecule and joins the nucleotides together, forming a phosphodiester bond b/w sugar and phosphate groups of nucleotides.

⇒ This process is repeated to create a chain of nucleotides, forming the backbone of DNA or RNA.

✓ How Psoralen is activated? (39) *

- ⇒ Psoralen becomes activated when exposed to ultraviolet A (UVA) radiation, leading to a (thermic) photochemical reaction.
- ⇒ In this activated state, Psoralen forms covalent cross-links with DNA, causing DNA damage and associated biological effects.
- (10)

Promotes / Oligomers / Multimers: (45) *

✓ Promoter: Initiates transcription by binding RNA polymerase & transcription factors.

Oligomer: Short polymer with few repeating units, common in small DNA or protein sequences.

Multimer: Formed by multiple copies of a molecule, creating larger str.s with repeated sub units.

✓ Structure of Proteins: (46) *

- 1- Primary: Linear amino acid sequence in a protein connected by peptide bonds.
- 2- Secondary: Local folding patterns like alpha helices and beta sheets, held by hydrogen bonds.
- ★ 3- Tertiary: Overall 3D folding, shaped by distant amino acid interactions (hydrogen bonds, disulphide bonds, etc.)
- 4- Quaternary: Organization of multiple protein subunits to form a functional complex.

Function of Nucleic Acid: (58) *

Nucleic acids, such as DNA and RNA, play crucial roles in storing and transmitting genetic information. They also participate in protein synthesis, regulation of gene expression, and various cellular processes essential for life and inheritance.